

The mathematical core of CBMR improvements

Selected derivations for discussion

Based on the expanded CBMR derivations

Why focus on the mathematics?

Goal

Replace large automatic-differentiation calculations with formulas that expose the likelihood curvature and its structure.

- Fit the same specified likelihood; derive its score and information.
- Exploit affine predictors, repeated patterns, and structured Hessian solves.
- Keep statistical assumptions visible when simplifying computation.

Scale motivating the work

On the 21-group diagnosis pool, the original Hessian autodiff intermediate was estimated at **28.7 GB** for 8,420 coefficients and 21,789 voxels.

The formulas below target this intermediate, not the intrinsic cost of every possible covariance output.

One predictor, two effect structures

For study i , voxel j , and a fixed offset o_{ij} :

$$\eta_{ij} = \log \mu_{ij}.$$

Globally constant (GC)

$$\eta_{ij} = o_{ij} + B_j^\top \xi_{g(i)} + z_i^\top \gamma.$$

The moderator effect γ_q is constant over voxels.

Spatially varying (SV)

$$\eta_{ij} = o_{ij} + \sum_{r=1}^R z_{ir} B_j^\top \beta_r.$$

The effect of moderator r at voxel j is $B_j^\top \beta_r$.

For an affine design row x_{ij} , the predictor is $\eta_{ij} = o_{ij} + x_{ij}^\top \theta$. Fixed dispersion and offsets are assumed in the regression derivatives that follow.

Closed-form information: scalar derivative to matrix

Let $\ell_{ij}(\eta_{ij})$ be a cell log-likelihood, with $s_{ij} = \partial \ell_{ij} / \partial \eta_{ij}$ and $w_{ij} = -\partial^2 \ell_{ij} / \partial \eta_{ij}^2$. Because η_{ij} is affine in θ :

$$U = \nabla_{\theta} \ell = \sum_{i,j} x_{ij} s_{ij}, \quad \boxed{H = -\nabla_{\theta}^2 \ell = \sum_{i,j} w_{ij} x_{ij} x_{ij}^{\top} .}$$

- No second derivative of the predictor appears; a nonlinear predictor would add a score-weighted curvature term.
- Poisson: $s = y - \mu$, $w = \mu$.
- Independent NB2: $s = (y - \mu)/(1 + \alpha\mu)$ and $w^{\text{obs}} = \mu(1 + \alpha y)/(1 + \alpha\mu)^2$.

Sign convention

H is observed information (negative log-likelihood curvature); $\nabla^2 \ell = -H$.

Poisson: exact compression without losing study allocation

Under a separable exposure, write $\mu_{ij} = u_{g(i)j}v_i$, where $u_{gj} = e^{B_j^\top \xi_g}$ and $v_i = e_i e^{z_i^\top \gamma}$. Define $Y_{gj} = \sum_{i \in \mathcal{I}_g} Y_{ij}$ and $n_i = \sum_j Y_{ij}$.

$$\ell = \sum_{g,j} Y_{gj} B_j^\top \xi_g + \sum_i n_i z_i^\top \gamma - \sum_g \left(\sum_j u_{gj} \right) \left(\sum_{i \in \mathcal{I}_g} v_i \right) + C.$$

- This is an exact rearrangement of the full Poisson likelihood.
- The compressed statistics include both voxel totals Y_{gj} and study totals n_i .
- Fitting voxel totals alone drops the conditional allocation term $\sum_i n_i \log(v_i / T_g)$ and therefore changes the global-covariate information.

Takeaway

Compression is exact because the omitted allocation structure is retained in the study totals; it is not simply “sum counts over studies.”

Independent NB: observed and Fisher curvature differ

For NB2 with mean $\mu = e^\eta$ and dispersion α :

$$\ell_\eta = \frac{y - \mu}{1 + \alpha\mu}, \quad -\ell_{\eta\eta} = \underbrace{\frac{\mu(1 + \alpha y)}{(1 + \alpha\mu)^2}}_{w^{\text{obs}}}.$$

Taking expectation at fixed μ, α gives

$$\underbrace{E(w^{\text{obs}})}_{w^{\text{F}}} = \frac{\mu}{1 + \alpha\mu}, \quad \text{since } E(Y) = \mu.$$

- Observed weights depend on the realized counts; Fisher weights do not.
- Both assemble through $H = X^\top \text{diag}(w)X$, but they answer different curvature questions.
- As $\alpha \rightarrow 0$, both weights tend to the Poisson weight μ .

Structural warning

For NB, separable means do not generally imply separable information weights; a Poisson Kronecker shortcut is not automatically valid.

Aggregated NB: moment matching is an approximation

Suppose independent cells have means $\mu_{ij} = u_j v_i$ and variance $\mu_{ij} + \alpha \mu_{ij}^2$. With $s_k = \sum_i v_i^k$:

$$m_j = E(Y_{.j}) = u_j s_1, \quad \text{Var}(Y_{.j}) = u_j s_1 + \alpha u_j^2 s_2.$$

Matching an NB variance $m_j + m_j^2/\rho$ yields

$$\rho = \frac{s_1^2}{\alpha s_2}, \quad A = \frac{s_1}{\alpha s_2}, \quad p_j = \frac{u_j}{u_j + A}.$$

- This matches the first two moments, not generally the distribution.
- The third-cumulant discrepancy is $2\alpha^2 u_j^3 (s_3 - s_2^2/s_1) \geq 0$.
- Equality holds for equal positive study factors; otherwise the convolution is generally not exactly NB.

Interpretation

Derivatives can be exact for this *working likelihood*, while the likelihood itself remains an approximation to the summed-cell model.

Aggregated NB: the composite Hessian has inner curvature

The working negative log-likelihood is first differentiated in coordinates (η, ρ, A) ; then ρ and A are composed with global coefficients γ . For a, b indexing components of γ :

$$\begin{aligned}\Psi_{\gamma_a \gamma_b} = & \Psi_{\rho\rho} \rho_a \rho_b + \Psi_{\rho A} (A_b \rho_a + \rho_b A_a) + \Psi_{AA} A_a A_b \\ & + \Psi_{\rho} \rho_{ab} + \Psi_A A_{ab}.\end{aligned}$$

- The last line is the curvature of the inner maps $\rho(\gamma), A(\gamma)$.
- It does not generally vanish at an observed regression optimum: Ψ_{ρ} and Ψ_A need not separately be zero.
- It vanishes in expectation under the correctly specified matched distribution, not simply by setting the observed score to zero.

Proof checkpoint

Differentiate the composite score $\Psi_{\rho} \rho_a + \Psi_A A_a$ once more; both product-rule terms produce the highlighted contribution.

Clustered NB: integrate one shared study effect

Let $\Lambda_i \sim \text{Gamma}(\nu, \text{rate } \nu)$ be shared by all voxels in study i , and $Y_{ij} \mid \Lambda_i \sim \text{Poisson}(\Lambda_i \mu_{ij})$. Writing $n_i = \sum_j y_{ij}$ and $t_i = \sum_j \mu_{ij}$, gamma integration gives

$$\ell_i = \log \Gamma(n_i + \nu) - \log \Gamma(\nu) + \nu \log \nu - (n_i + \nu) \log(t_i + \nu) + \sum_j y_{ij} \eta_{ij} + C(y_i).$$

For $\boldsymbol{\mu}_i = (\mu_{i1}, \dots, \mu_{iN})^\top$ and $c_i = (n_i + \nu)/(t_i + \nu)$, the observed information in the log means is

$$W_i^{\text{obs}} = c_i \left\{ \text{diag}(\boldsymbol{\mu}_i) - \frac{\boldsymbol{\mu}_i \boldsymbol{\mu}_i^\top}{t_i + \nu} \right\}.$$

- Integrating a shared multiplier induces off-diagonal dependence.
- The negative rank-one term is essential; independent NB diagonal weights describe a different model.
- For an affine study design X_i , project directly: $H_i = X_i^\top W_i^{\text{obs}} X_i$.

Schur complement: shared effects couple otherwise separate maps

Disjoint spatial design components give $D = \text{blkdiag}(D_1, \dots, D_G)$, but shared global coefficients supply a border:

$$H = \begin{pmatrix} D & F \\ F^\top & \mathcal{E} \end{pmatrix}, \quad T = D^{-1}F, \quad S = \mathcal{E} - F^\top T.$$

From $Dx_D + Fx_\gamma = b_D$, substitute $x_D = D^{-1}b_D - Tx_\gamma$ into the second block equation:

$$Sx_\gamma = b_\gamma - F^\top D^{-1}b_D, \quad x_\gamma = S^{-1}(b_\gamma - F^\top D^{-1}b_D), \quad x_D = D^{-1}b_D - Tx_\gamma.$$

The inversion step

Factor the smaller D_g blocks and the $Q \times Q$ Schur matrix S ; solve against right-hand sides rather than explicitly inverting H .

The displayed elimination requires invertible D and S . For an SPD H both are SPD; an indefinite or singular system needs appropriate pivoting or may have no ordinary inverse.

What the block inverse proves about uncertainty

Expanding the Schur solve in b_D and b_γ yields

$$H^{-1} = \begin{pmatrix} D^{-1} + TS^{-1}T^\top & -TS^{-1} \\ -S^{-1}T^\top & S^{-1} \end{pmatrix}.$$

For different groups $g \neq h$, $\text{Cov}(\hat{\xi}_g, \hat{\xi}_h) \approx T_g S^{-1} T_h^\top$ under the model-based covariance approximation.

- Although D_g and D_h do not interact in H , their inverse cross block generally is nonzero.
- The correction $TS^{-1}T^\top$ has rank at most Q ; when H is SPD it is positive semidefinite.

Small check

For $D = \text{diag}(2, 3)$, $F = (1, 1)^\top$, $\mathcal{E} = 2$: $T = (1/2, 1/3)^\top$, $S = 7/6$, so the cross-group inverse entry is $(1/2)(6/7)(1/3) = 1/7$, not zero.

The inverse identity is exact algebra; interpreting it as an estimator covariance is a local/asymptotic step.

Inversion for contrasts: never form an unwanted full covariance

For k requested linear contrasts (rows of C), solve

$$HX = C^\top, \quad \boxed{V_C = CX = CH^{-1}C^\top}.$$

Only k right-hand sides are needed. Partition $C = [C_D \ C_\gamma]$ and put $A_C = C_\gamma - C_D T$. The block inverse gives the smaller calculation

$$\boxed{V_C = C_D D^{-1} C_D^\top + A_C S^{-1} A_C^\top}.$$

- Factor each D_g and S once; compute both terms by solves.
- For a spatial-only contrast, $C_\gamma = 0$ but the Schur term remains unless $C_D T = 0$.
- Factor storage: $O(\sum_g d_g^2 + dQ + Q^2)$ rather than $O((d + Q)^2)$ for dense factors; a full covariance output still takes $O((d + Q)^2)$ entries.

Factorization work after assembly: $O(\sum_g d_g^3 + Q \sum_g d_g^2 + dQ^2 + Q^3)$ versus $O((d + Q)^3)$ dense; savings depend on actual block sizes.

Shared spatial patterns: one local calculation, many blocks

If pattern h has spatial loading $L_h \in \mathbb{R}^R$, its effective map is $c_h = J_h \beta$, with $J_h = L_h^\top \otimes I_P$. For pattern curvature $K_h = \nabla_{c_h}^2 F_h$:

$$H_{\beta\beta} = \sum_h J_h^\top K_h J_h = \sum_h (L_h L_h^\top) \otimes K_h.$$

- Compute local pattern curvature once, then scatter it into all coefficient blocks (including cross-terms for overlapping loadings).
- For independent cells, $B^\top \text{diag}(w) B$ can be formed by row-weighting B rather than constructing $\text{diag}(w)$.
- This avoids a parameter-by-pattern-by-voxel autodiff intermediate.

Exact boundary

The information may have exploitable block or pattern structure, but a requested dense covariance still requires $O(n_\theta^2)$ storage.

Checks run: symbolic identities and numerical implementations

Exact symbolic checks

- make proofs: **120 claims closed.**
- Every accepted SymPy residual reduced exactly to zero.

Implementation tests

- make numerical: **29 tests passed.**
- Comparison unit tests: **13 tests passed.**

Seeded floating-point regression

- **512 required checks; 0 failures.**
- Derivatives: 312 checks; largest relative Frobenius error 3.14×10^{-12} .
- Covariance: 29 checks; largest relative error 2.99×10^{-16} .
- Fit agreement: 100 checks; largest relative error 3.47×10^{-6} .
- 47 additional diagnostics retained for difficult cases; they are reported separately from required failures.

Float64 derivatives were checked against independent autodiff; structured covariance against dense references; fits at matched objectives. Dispersion was fixed.

Formal coverage and measured full-pool performance

Lean/Mathlib audit

- **358 theorem declarations audited**; the axiom audit passed.
- Appendix display inventory: **30 proved, 15 partial, 9 pending** (54 total).
- Prose derivations: **8 proved, 1 partial, 6 pending** (15 total).
- Thus the Lean check is substantive but the full appendix is **not fully formalised**.

Real 21-group diagnosis pool

Same full run, including 20 one-versus-rest contrasts:

Method	Time	Peak memory
Stock autodiff	did not run	28.7 GB estimated
Chunked autodiff	769 s	5.49 GB
Closed form	44 s	2.67 GB

One measured run, not a performance guarantee. Lean formalisation is partial; empirical agreement is not a formal proof.

What is proved, and what remains a modeling choice

Evidence in the repository

- SymPy checks exact symbolic identities for selected derivatives and matrix formulas.
- Lean/Mathlib checks a subset of the derivations.
- Independent numerical tests compare derivatives, solves, and fits.

Keep these distinctions

- Exact differentiation does not make an approximate likelihood exact.
- Inverse information is not an exact finite-sample estimator variance.
- Mathematical formulas do not imply every model variant is implemented.
- Numerical agreement is not formal proof or inferential calibration.

Primary source: `docs/cbmr_expanded_appendix.tex` and `docs/expanded_covariance.tex`. The current Lean inventory explicitly does not certify every handwritten intermediate derivation.

Take-home points

- ❶ The affine log-link turns scalar likelihood curvature into a transparent weighted design cross-product.
- ❷ Correct reductions depend on the likelihood: Poisson compression is exact with study totals; aggregated NB is moment matched; clustered NB retains shared-study dependence.
- ❸ Block and Schur algebra avoid a full Hessian inverse for selected contrasts while retaining uncertainty from shared effects.

Suggested discussion

Which distinction is most consequential for the scientific analysis: GC versus SV effects, the choice among NB constructions, or joint uncertainty from shared global covariates?